

Negative results and claim revision

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Robustness across conditions

Without opening the old lesson, explain one key idea from Robustness across conditions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Running the primary experiment

Without opening the old lesson, explain one key idea from Running the primary experiment. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Pilot experiments

Without opening the old lesson, explain one key idea from Pilot experiments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Negative results and claim revision. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Negative results and claim revision. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Hypothesis predicts mean latency reduction at least 2 seconds. Observed paired mean reduction .8. Report.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Hypothesis expects B support strictly above A. Both pass 8/10 reserved cases. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: Failure to meet a criterion or reject a null is not proof of exact equivalence. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Hypothesis predicts mean latency reduction at least 2 seconds. Observed paired mean reduction .8. Report.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled negative results and claim revision evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of negative results and claim revision. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

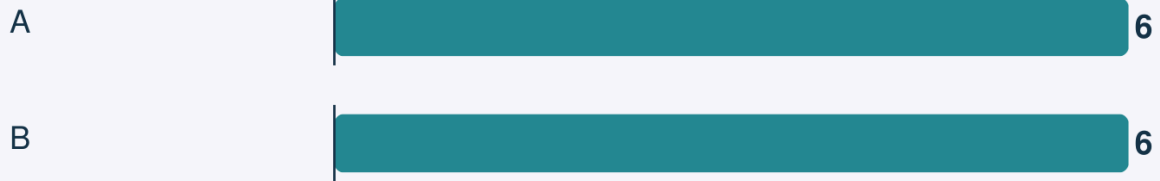
12. Pictorial case practice: Negative results and claim revision

A negative result changes the claim. A proposed prompt improvement ties the baseline on a fixed eight-case evaluation.

A and B each score 6/8. Before running, the team predicted B would score higher.

No observed count advantage

correct cases out of 8 shared cases



Bar length shows the supplied value; the marked baseline is zero.

Negative results and claim revision

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Recall the prediction

The claim was a higher correct count for B.



2. State the finding

Both versions score 75%; no observed count advantage appears here.



3. Revise cautiously

Report the unsupported prediction and a justified next test.

CHANGE ONE THING

B wins on one new task after the team chooses it because A fails.

Explain the changed result using the matching lesson.

Does a 6/8 tie support the prediction that B scores higher? Explain why one newly selected favourable demonstration does not replace the original result or prove equivalence.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A negative result changes the claim. A proposed prompt improvement ties the baseline on a fixed eight-case evaluation.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Report the unsupported prediction and a justified next test.

1. Revise cautiously
2. Recall the prediction
3. State the finding

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Revise cautiously Report the unsupported prediction and a justified next test. A negative result can be informative when methods and limits are clear. A tie does not prove the methods are equivalent under all conditions or that the experiment was worthless.

Practice 2: explain the deciding evidence

Changed-case practice: B wins on one new task after the team chooses it because A fails. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. Treat that selected demonstration as exploratory, not a replacement for the original result.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Treat that selected demonstration as exploratory, not a replacement for the original result. Case selection now depends on outcomes.

Repair a missing building block

Review Robustness across conditions (ai-7/27). Goal: Explain the mechanism and terminology of robustness across conditions. Short practice: Calculate indoor, outdoor and combined rates. Which relevant weakness disappears from view when only the combined rate is reported?. Return to this week and explain what you changed.

Review Running the primary experiment (ai-7/25). Goal: Explain the mechanism and terminology of running the primary experiment. Short practice: Compute A's and B's six-case counts and identify the cases that favour each. How must a B revision tuned on case four be labelled when evaluated again on the same set?. Return to this week and explain what you changed.

Review Pilot experiments (ai-7/22). Goal: Explain the mechanism and terminology of pilot experiments. Short practice: Calculate reviewer agreement on the four pilot answers. What can the disagreement improve, and why are those revised practice answers not untouched final evidence?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

